

Assessment of β -Variational Auto Encoder (VAE) for Airfoil and Cylinder Flows

Srijan Satyal and Nek Sharan

May 12, 2026

1 Introduction

The purpose of this report is to compare the reduced-order representation obtained from β -VAE with the classical POD approach. The same framework is applied to cylinder and airfoil flow datasets. The analysis follows the reduced-order modeling idea used by Solera-Rico et al. [1], where the β -VAE was shown to provide compact nonlinear latent representations for fluid-flow reconstruction and prediction.

The fluctuating velocity field is represented using the streamwise and crosswise velocity components,

$$\mathbf{q}(t) = [u(x, y, t), v(x, y, t)]. \quad (1)$$

The β -VAE maps each flow snapshot into a low-dimensional latent space via encoder and reconstructs the velocity field through the decoder. POD is used as the linear reference method. The reconstruction quality is measured using cumulative reconstructed energy,

$$E = \left[1 - \frac{\left\langle \sum_{i=1}^{N_p} \sum_{j=1}^2 (q_{ij} - \tilde{q}_{ij})^2 \right\rangle}{\left\langle \sum_{i=1}^{N_p} \sum_{j=1}^2 q_{ij}^2 \right\rangle} \right] \times 100\%, \quad (2)$$

where q_{ij} is the reference velocity fluctuation, $\tilde{q}_{ij}$ is the reconstructed value, and $\langle \cdot \rangle$ denotes averaging over snapshots.

For prediction, the trained β -VAE encoder first maps the flow field into latent space. The easy-attention model then predicts the future latent variables, and the decoder reconstructs the predicted velocity field. Therefore, the prediction is evaluated both in latent space and in the reconstructed physical field.

2 Circular-cylinder Flow

The circular-cylinder case is a canonical vortex-shedding problem. The wake is more periodic than the separated airfoil case, so the dominant behavior is expected to be low-dimensional. The β -VAE energy and mode results show that two latent variables are sufficient to represent the primary shedding dynamics. The first two latent variables contain a clear dominant frequency at $f/f_c \approx 0.045$, and their temporal signals are smooth and periodic.

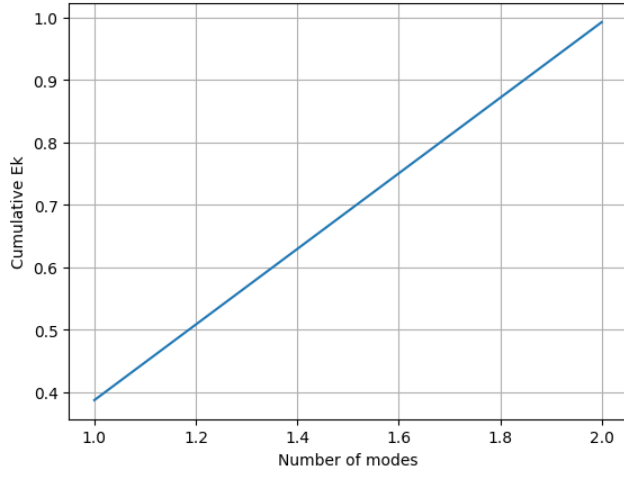
The POD result also shows the same behavior. The first POD mode captures approximately 49% of the energy, and the first two modes capture about 95%. Higher modes represent harmonic content and smaller corrections. The first two POD modes have the same dominant frequency as the β -VAE modes, while modes 3–4 and 5–6 correspond to higher harmonics near $f/f_c \approx 0.090$ and 0.135 .

The cylinder prediction results are shown in Fig. 3. At both $t + 65t_c$ and $t + 100t_c$ ($t_c = dt$ obtained from snapshot sampling time), the predicted fields agree closely with the true and reconstructed fields. The streamwise velocity captures the main low-speed wake region behind the cylinder, while the crosswise velocity captures the alternating vortex-shedding structure. The prediction remains accurate at long prediction time because the cylinder wake is governed by a smooth periodic latent-space trajectory.

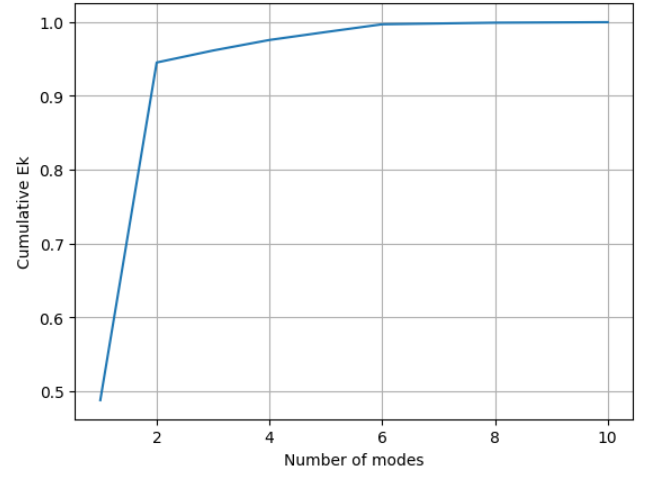
3 Airfoil at angle of attack (α) = 10°

The airfoil case at $\alpha = 10^\circ$ represents a relatively organized wake compared with the higher-angle case. The β -VAE cumulative reconstructed energy is shown in Fig. 4. The energy increases from about 40% with one latent variable to almost 99% with two latent variables. This indicates that the dominant unsteady behavior of this flow can be represented using only two latent variables.

For POD, the first mode captures about 48% of the energy, while the first two modes capture approximately 95%. The energy approaches nearly 99% after about four modes. Therefore, POD also performs well for this case, but it requires more modes than the β -VAE to reach the same reconstruction level. This is expected because the flow is periodic and can be represented efficiently by mode pairs.

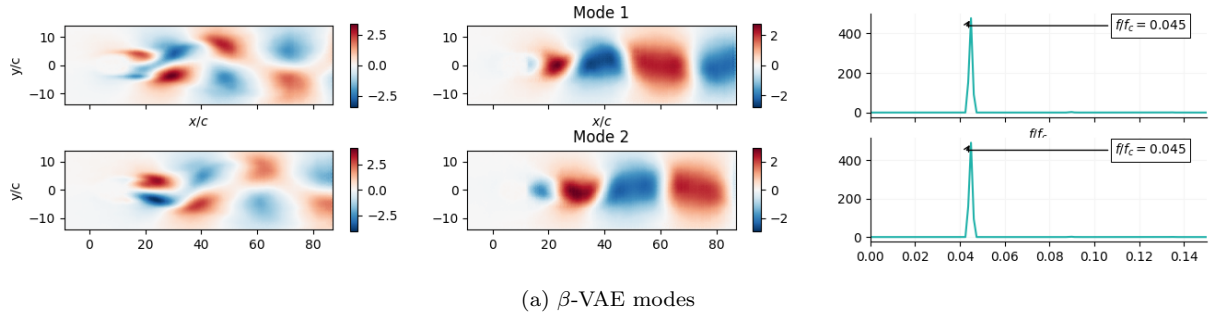


(a) β -VAE

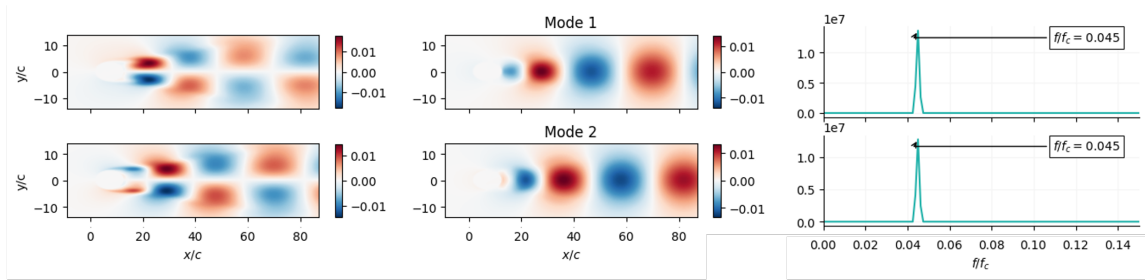


(b) POD

Figure 1: Cumulative reconstructed energy for the cylinder.



(a) β -VAE modes



(b) POD modes

Figure 2: Spatial modes and spectra for the circular-cylinder flow.

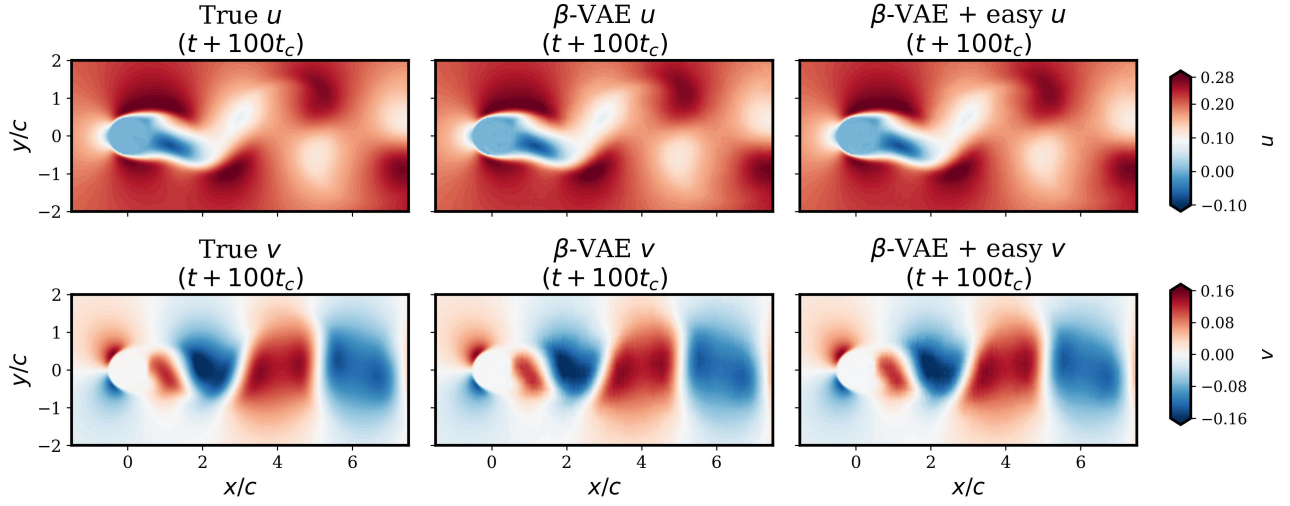
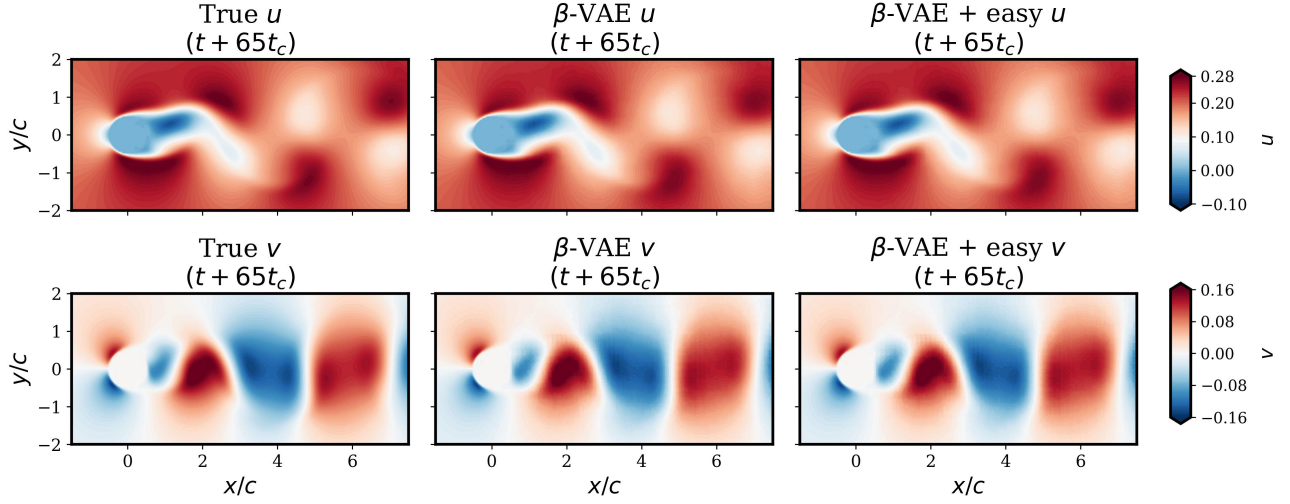


Figure 3: True, reconstructed, and predicted fields for the circular-cylinder flow.

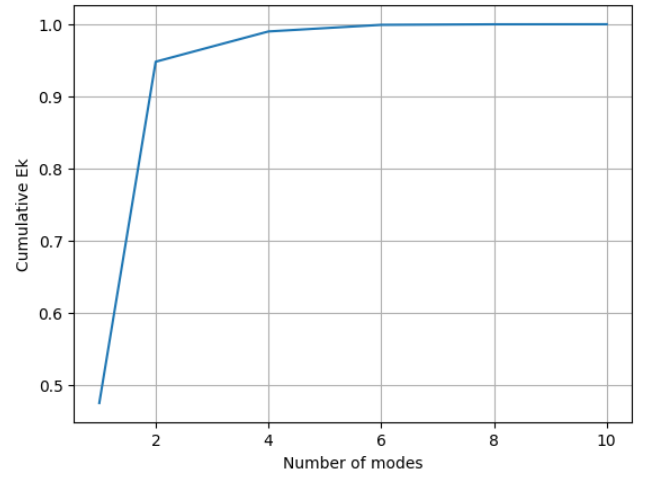
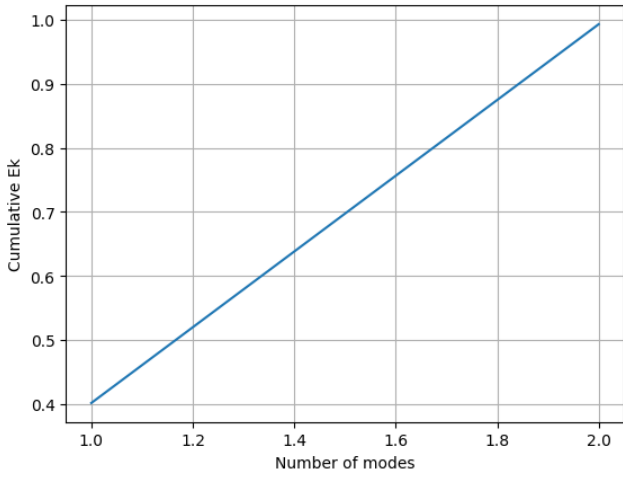


Figure 4: Cumulative reconstructed energy for the airfoil case at $\alpha = 10^\circ$.

The β -VAE modes are shown in Fig. 5. The two modes show alternating positive and negative wake structures downstream of the airfoil. The spectra show a clear dominant frequency at $f/f_c \approx 0.038$ for both latent variables. This means that the two latent variables represent the same wake-shedding process with a phase shift.

The POD modes also identify the same dominant shedding frequency. The first two POD modes have a peak at $f/f_c \approx 0.038$, while higher POD mode pairs contain harmonic frequencies near $f/f_c \approx 0.078$ and 0.116 . This confirms that both POD and β -VAE capture the same dominant physics. However, the β -VAE represents the primary wake dynamics more compactly.

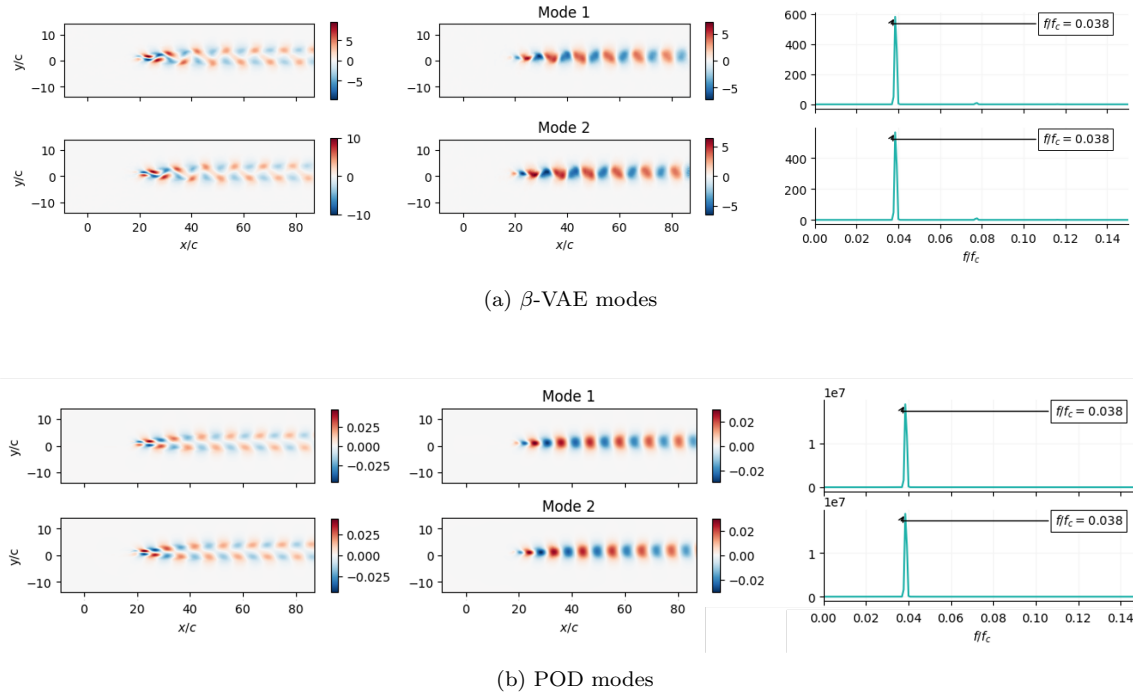


Figure 5: Spatial modes and spectra for the airfoil case at $\alpha = 10^\circ$.

The prediction result for the $\alpha = 10^\circ$ case is shown in Fig. 6. The predicted field from the β -VAE plus easy-attention model closely follows both the true field and the direct β -VAE reconstruction. The streamwise velocity preserves the low-speed wake region and the downstream alternating structures. The crosswise velocity also captures the sign and location of the main vortical structures.

4 Airfoil at angle of attack (α) = 35°

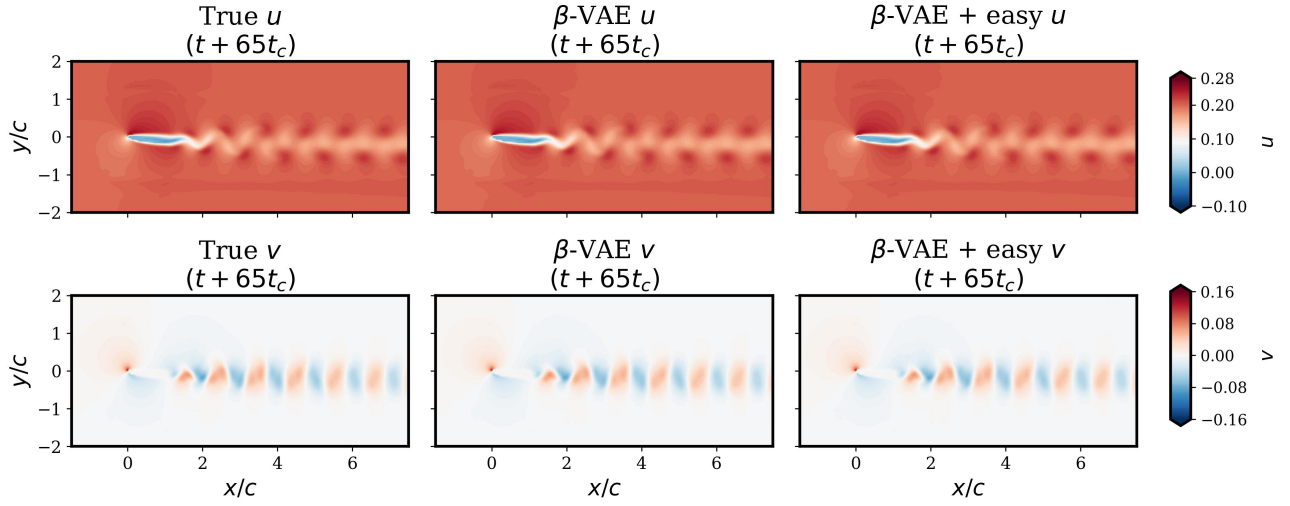
The airfoil case at angle of attack (α) = 35° is more challenging because the flow contains stronger separation and more nonlinear wake dynamics. The β -VAE energy curve in Fig. 7 shows that approximately five latent variables reconstruct about 74% of the flow-field energy, while about eight latent variables reconstruct nearly the full energy. After this point, additional modes contribute very little. This shows that the β -VAE still provides a compact representation even for a separated and nonlinear wake.

The POD energy curve increases more slowly. With ten POD modes, the reconstructed energy is only about 82%. This shows that POD requires more modes to represent the separated airfoil wake. The difference occurs because POD is a linear decomposition, so nonlinear wake deformation and vortex interaction are distributed over many modes. In contrast, the β -VAE decoder reconstructs the flow through a nonlinear mapping.

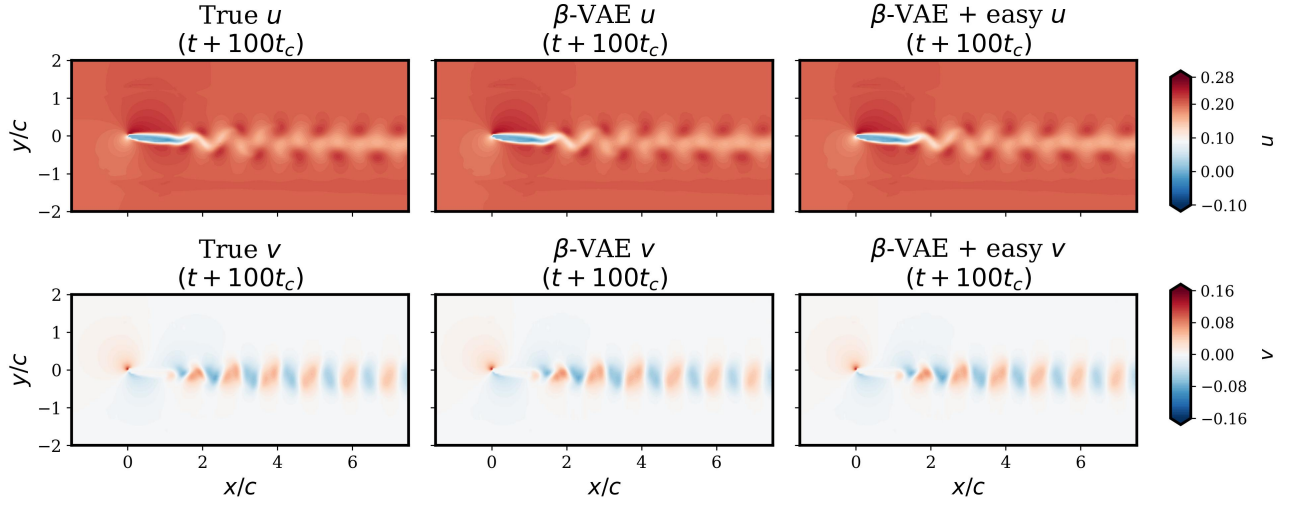
The β -VAE and POD modes are compared in Fig. 8. The β -VAE modes contain large-scale coherent structures in the separated wake. The dominant frequency content is concentrated at low nondimensional frequencies, mainly around $f/f_c \approx 0.003$ – 0.004 . These modes represent large-scale, low-frequency separated-flow motion.

The POD modes also capture coherent wake structures, but the flow information is spread over more modes. Some modes represent large-scale wake motion, while higher modes act as corrections to reconstruct the separated flow. This is consistent with the slower growth of the POD energy curve.

The prediction result for the $\alpha = 35^\circ$ case is shown in Fig. 9. The direct β -VAE reconstruction agrees well with the true field, showing that the decoder can reconstruct the separated wake from the latent variables. The predicted field also preserves the main wake structures.

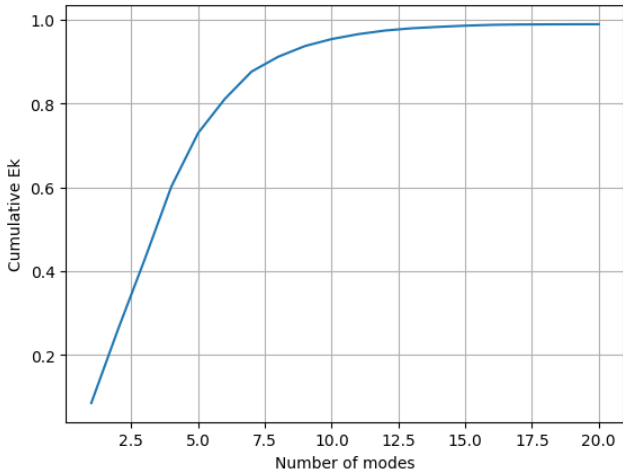


(a) $t + 65t_c$

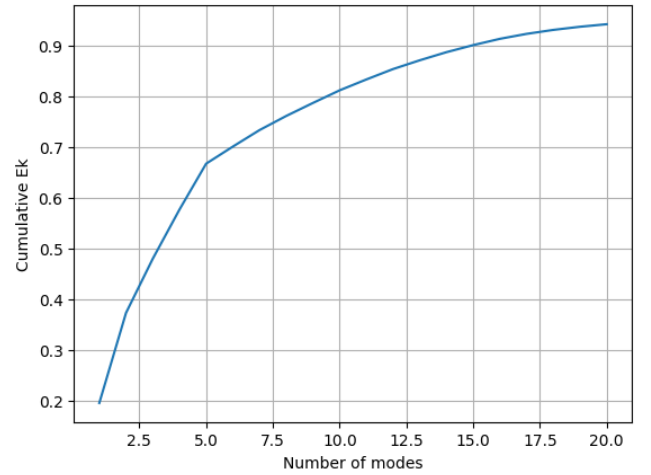


(b) $t + 100t_c$

Figure 6: True, reconstructed, and predicted fields for the airfoil case at $\alpha = 10^\circ$.

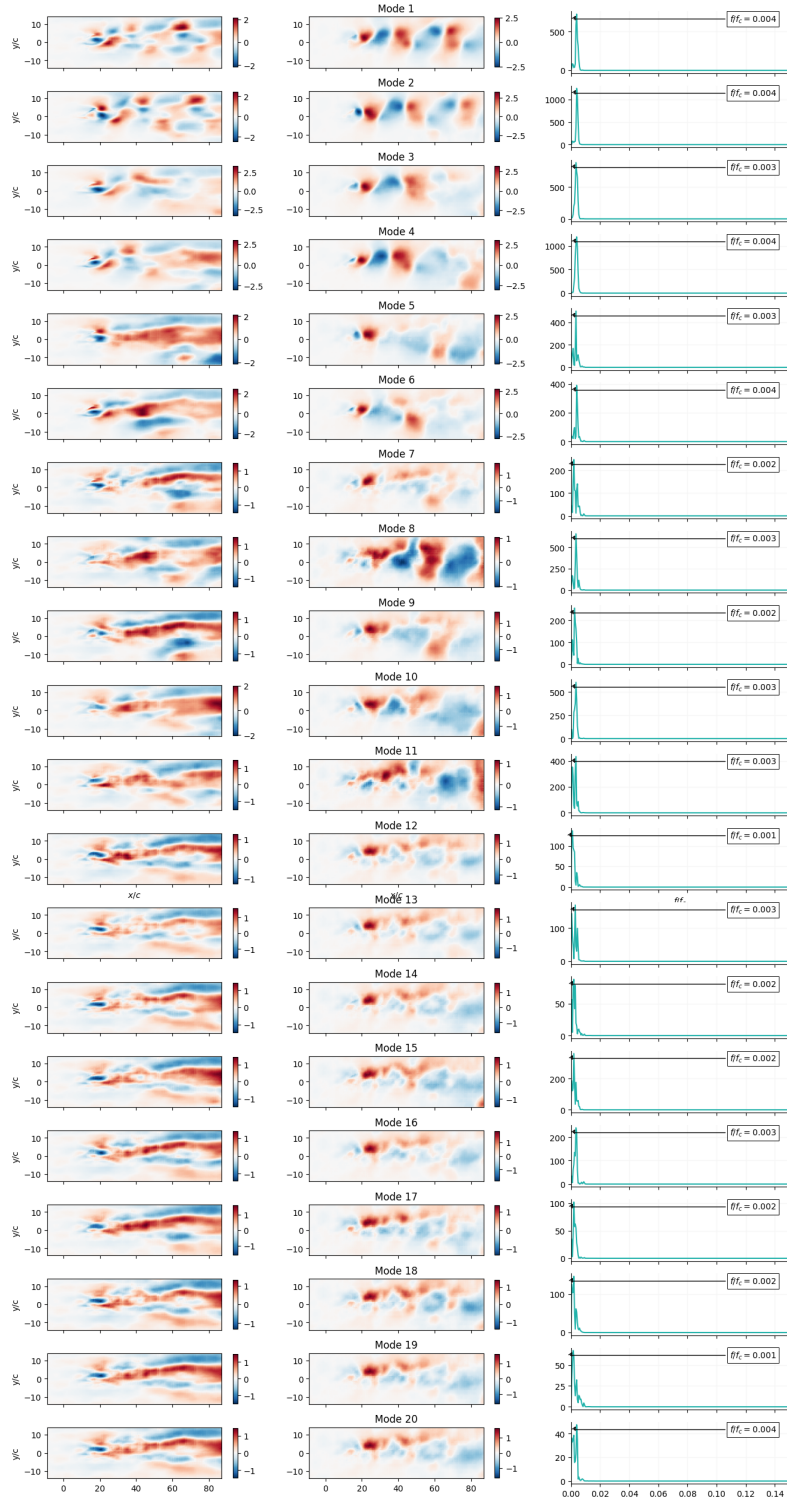


(a) β -VAE

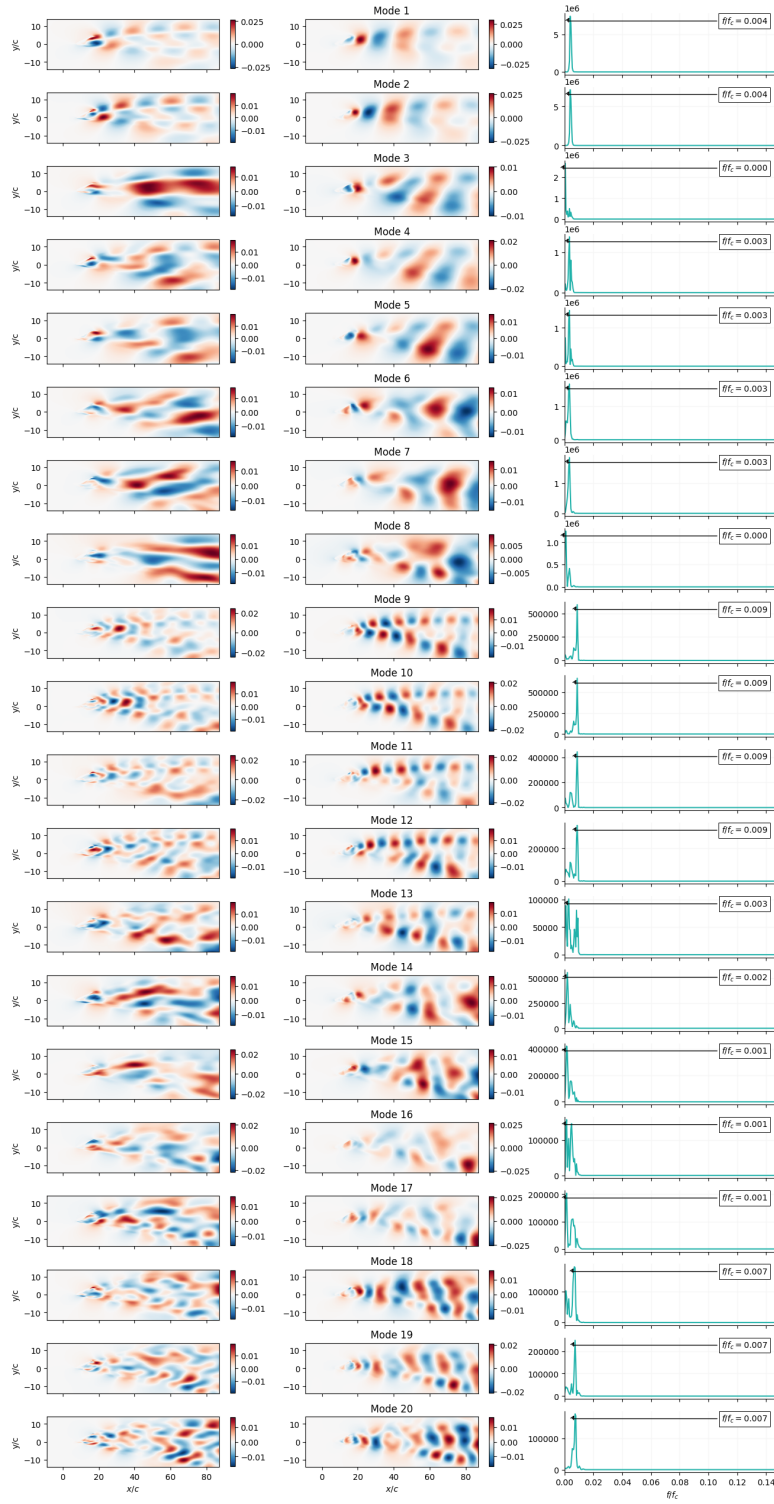


(b) POD

Figure 7: Cumulative reconstructed energy for the airfoil case at $\alpha = 35^\circ$.

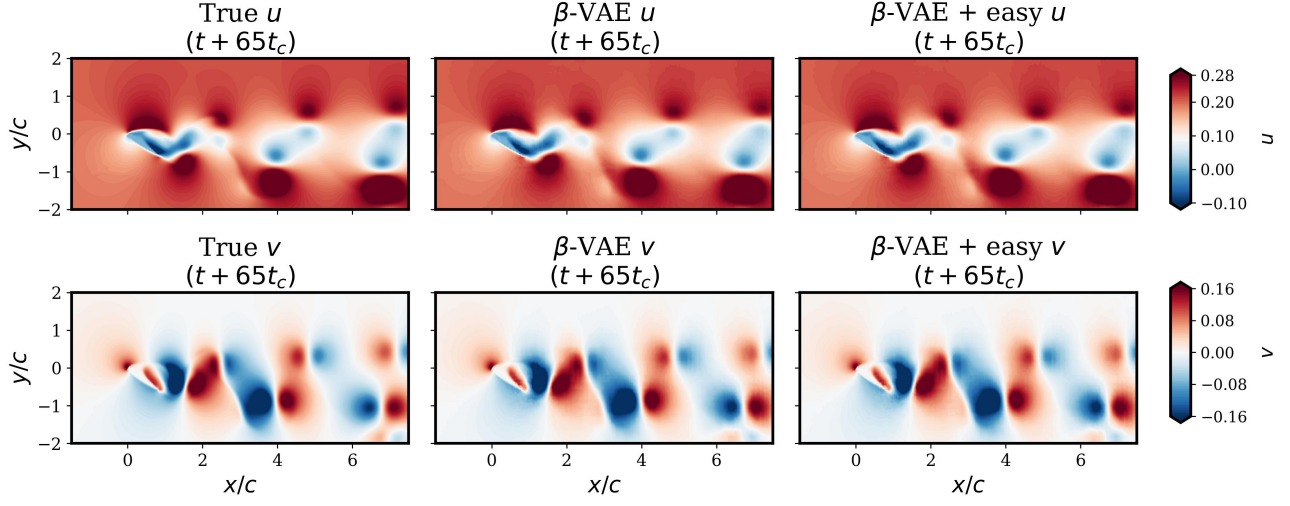


(a) β -VAE modes

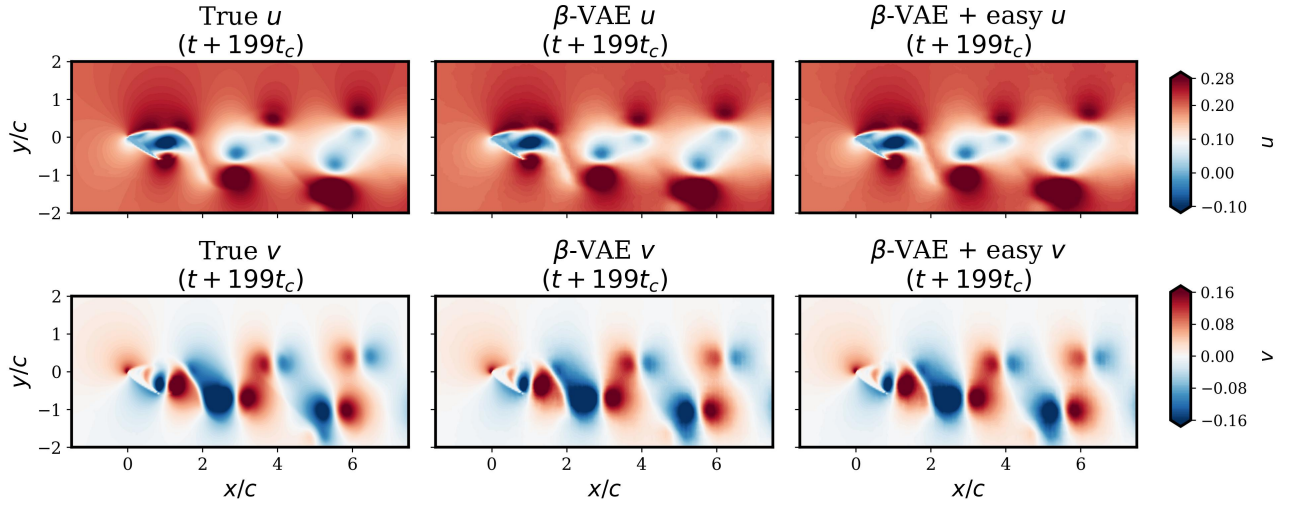


(b) POD modes

Figure 8: Spatial modes and spectra for the airfoil case at $\alpha = 35^\circ$.



(a) $t + 65t_c$



(b) $t + 199t_c$

Figure 9: True, reconstructed, and predicted fields for the airfoil case at $\alpha = 35^\circ$.

5 Comparison and Conclusions

The results show that the β -VAE provides a compact nonlinear representation for all three cases. For the airfoil at $\alpha = 10^\circ$, the wake is dominated by organized periodic motion. The β -VAE reconstructs almost the full energy using two latent variables, while POD requires more modes to reach the same level. For the cylinder case, both POD and β -VAE perform well because the wake is strongly periodic. The first two POD modes capture about 95% of the energy, and the β -VAE also learns a two-dimensional periodic latent representation.

The clearest advantage of the β -VAE appears for the airfoil at $\alpha = 35^\circ$. In this case, the flow is strongly separated and nonlinear. The β -VAE reaches nearly full reconstruction using about eight latent variables, while POD reaches only about 82% using ten modes. This shows that the nonlinear decoder is more efficient for representing separated wake structures.

Table 1: Summary of reduced-order representation behavior.

Case	Flow behavior	β -VAE result	POD result
Cylinder	Periodic vortex shedding	2 dominant modes at $f/f_c \approx 0.045$	First 2 modes capture about 95% energy
Airfoil $\alpha = 10^\circ$	Periodic wake	$\sim 99\%$ energy with 2 modes	$\sim 95\%$ with 2 modes; near 99% with 4 modes
Airfoil $\alpha = 35^\circ$	Strong separated wake	Nearly full energy with ~ 8 modes	About 82% with 10 modes

Overall, the β -VAE is a strong nonlinear alternative to POD for reduced-order representation of unsteady flow fields. POD performs well for periodic flows, but it becomes less compact when the flow contains stronger separation and nonlinear wake dynamics. The easy-attention model can also predict the learned latent dynamics. The prediction is fairly accurate for the cylinder and $\alpha = 10^\circ$ cases and $\alpha = 35^\circ$ case predicting the main coherent wake structures.

References

- [1] Alberto Solera-Rico, Carlos Sanmiguel Vila, Miguel Gómez-López, Yuning Wang, Abdulrahman Almashjary, Scott TM Dawson, and Ricardo Vinuesa. β -variational autoencoders and transformers for reduced-order modelling of fluid flows. *Nature Communications*, 15(1):1361, 2024.